Hausdorff dimension of self-similar sets in complete metric spaces

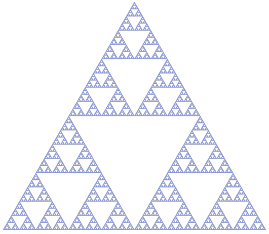
Barnabas Tanczos
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University of St Andrews

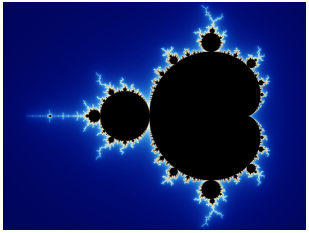
October 14, 2026

Overview

1. Introduction
2. Hausdorff dimension
3. Iterated function systems
4. Self-similar sets

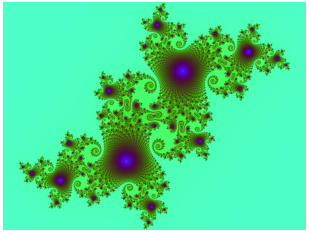
Introduction

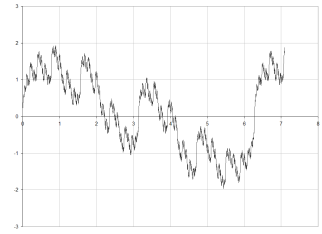














Benoit Mandelbrot (1924 - 2010)

- Polish mathematician
- pioneered fractal geometry in *Fractals: Form, Chance and Dimension*

Introduction

What are fractals?

- self-similar
- formed by (infinite) iterative process
- detail at arbitrary scales
- too irregular to describe using traditional methods
- properties that do not reflect topological dimension

Introduction

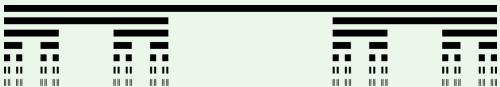
Example (Cantor ternary set)

$C_0 = [0, 1]$

$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$

$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{3}{9}\right] \cup \left[\frac{6}{9}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$

...



Example (Cantor ternary set)

It is ‘small’:

- length 0
- topological dimension 0

But also ‘large’:

- uncountable
- no isolated points

Hausdorff outer measure

Definition (Hausdorff outer measure)

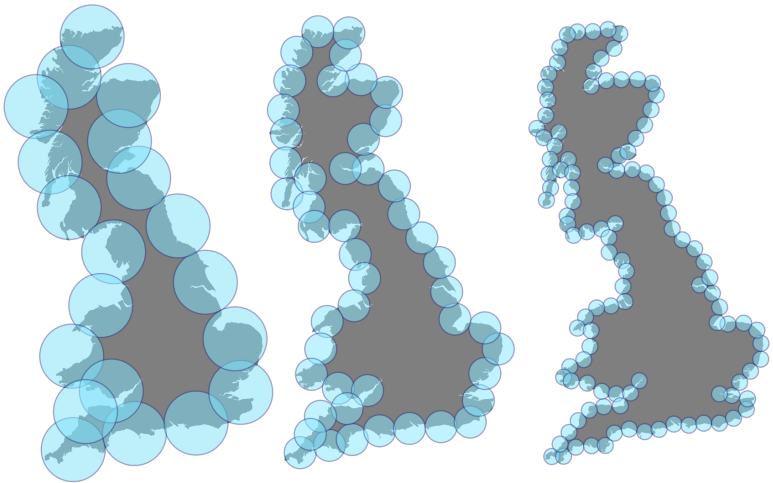
Let (X, d) be a metric space and let $S \subseteq X$. For $\delta > 0$ and $s \geq 0$ define

$$\mathcal{H}^s_\delta(S) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s : S \subseteq \bigcup_{i=1}^{\infty} U_i \text{ and } |U_i| < \delta \right\}.$$

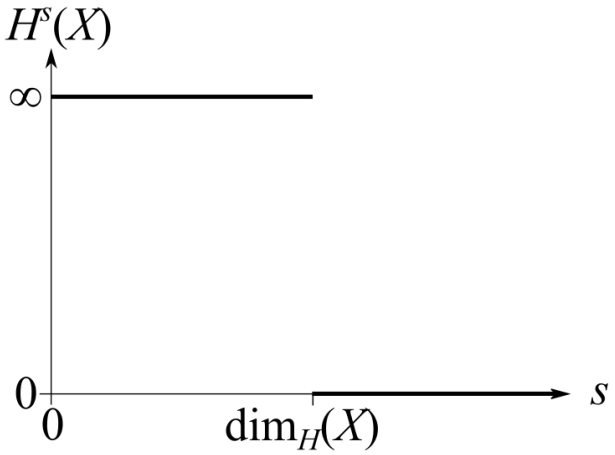
Then the s-dimensional **Hausdorff outer measure** is

$$\mathcal{H}^s(S) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(S).$$

Hausdorff outer measure



Hausdorff dimension



Hausdorff dimension

Definition (Hausdorff dimension)

The **Hausdorff dimension** of a set S is given by

$$\dim_H S = \inf\{s \geq 0 : \mathcal{H}^s(S) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(S) = \infty\}.$$

Hausdorff dimension

Theorem

Let A be a countable subset of X. Then $\dim_H A = 0$.

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Theorem

Let $\{A_n\}$ be subsets of X. Then $\dim_H \left(\bigcup_n A_n\right) = \sup_n \{\dim_H A_n\}$.

Hausdorff dimension

Example (Cantor ternary set)

The Cantor ternary set has Hausdorff dimension $s = \log 2 / \log 3$.

Contractions

Definition (Contraction)

A function $f : X \to X$ is a **contraction** on a metric space (X, d) if there is a ratio $0 \leq c < 1$ such that

$$d(f(x), f(y)) \leq cd(x, y)$$

for all $x, y \in X$.

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Theorem (Banach fixed-point theorem)

Let f be a contraction on a non-empty complete metric space (X, d). Then f has a unique fixed point $x_0 \in X$ where $f(x_0) = x_0$.

Iterated function systems

Definition (Iterated function system)

An **iterated function system (IFS)** is a finite family of contractions $\{f_1, \ldots, f_n\}$.

Iterated function systems

Definition (Iterated function system)

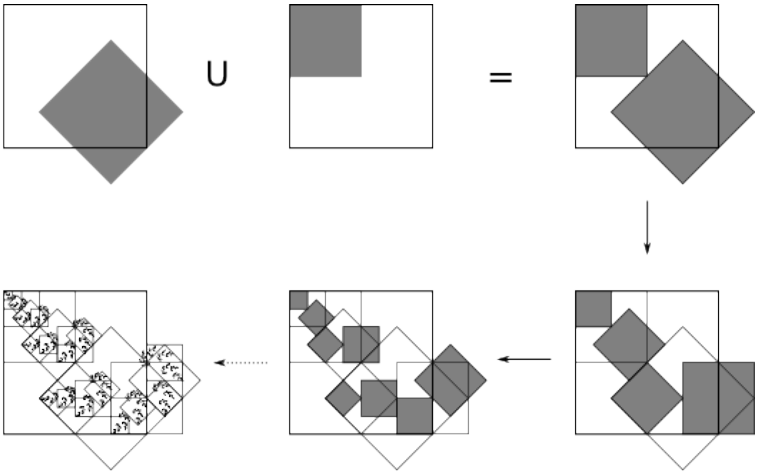
An **iterated function system (IFS)** is a finite family of contractions $\{f_1, \ldots, f_n\}$.

Definition (Attractor)

A set S is called the **attractor** for an IFS $\{f_1, \ldots, f_n\}$ if

$$S = f(S) = \bigcup_{i=1}^{n} f_i(S).$$

Iterated function systems



Iterated function systems

Theorem (Unique attractor)

Let (X, d) be a non-empty complete metric space. An iterated function system $\{f_1, \ldots, f_n\}$ on X has a unique non-empty compact attractor S.

Iterated function systems

Example (Cantor ternary set)

Define the contractions $f_1, f_2 : [0,1] \to [0,1]$ by

$$f_1(x) = \frac{1}{3}x \quad \text{and} \quad f_2(x) = \frac{1}{3}x + \frac{2}{3}.$$

The Cantor ternary set satisfies

$$C = f_1(C) \cup f_2(C).$$

Self-similar sets

Definition (Similarity)

A function $f : X \to X$ is a **similarity** on a metric space (X, d) if there is a ratio $0 \leq c < 1$ such that

$$d(f(x), f(y)) = cd(x, y)$$

for all $x, y \in X$.

Self-similar sets

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Definition (Self-similar set)

A **self-similar set** is the attractor of an IFS consisting of similarities.

Self-similar sets

Definition (Similarity value)

Let $\{f_1, \ldots, f_n\}$ be an IFS of similarities with ratios $c_1, \ldots, c_n$. The **similarity value** s is given by

$$\sum_{i=1}^{n} c_i^s = 1.$$

Self-similar sets

Definition (Doubling space)

A metric space is **doubling** if there is a constant C such that any ball can be covered by C balls of half the radius.

Self-similar sets

Definition (Doubling space)

A metric space is **doubling** if there is a constant C such that any ball can be covered by C balls of half the radius.

Theorem (OSC dimension theorem)

Let $\{f_1, \ldots, f_n\}$ be an IFS on a doubling space with self-similar set S and similarity value s. If there is a non-empty bounded open set V such that

$$\bigcup_{i=1}^{n} f_i(V) \subseteq V$$

with the union pairwise disjoint, then $\dim_H S = s$.

Self-similar sets

Example (Cantor ternary set)

The Cantor ternary set fulfils the above conditions:

- $\mathbb{R}$ is doubling
- $(0, 1)$ is an open set

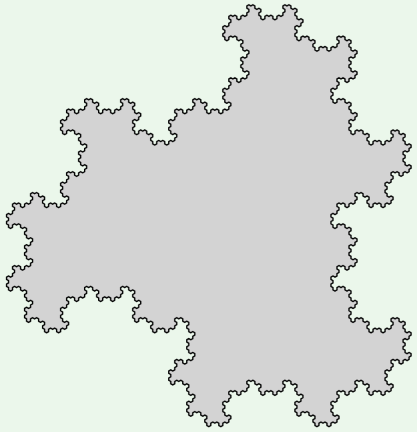
This implies the dimension is the solution of

$$2\left(\frac{1}{3}\right)^s = 1$$

and so $\dim_H C = \log 2 / \log 3$.

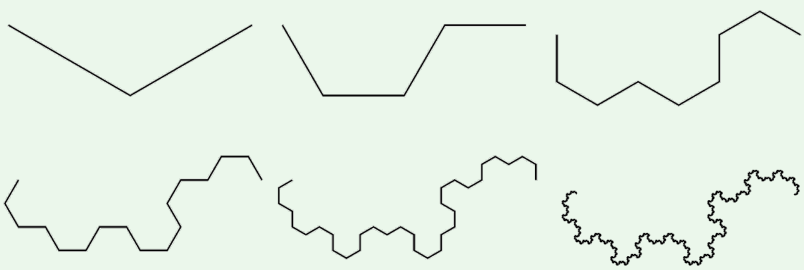
Self-similar sets

Example (Fudgeflake boundary)



Self-similar sets

Example (Fudgeflake boundary)



Questions

Any questions?

Thank you for your attention!
Feel free to ask any questions.

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